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Real-time ball-position measurement using multi-view cameras for live football broadcast

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Abstract We developed a robust ball-position-measurement system that can be used for actual football games. It measures the positions of a ball by integrating tracking results from multi-view cameras in real time. The positions can be used for calculating various statistical data and the production of live football broadcasts. Although several systems for visual object tracking have been proposed, tracking a ball during entire football games is still challenging due to many difficulties such as occlusion and illumination changes. We focused on the motion features of a ball object in addition to its appearance features. Tracking results from multi-view cameras are integrated in consideration of the reliability of these results, and the final ball position is robustly measured. The system was evaluated with videos shot during four professional football games held at multiple stadiums during the day and night. The evaluation results and an experimental demonstration indicate that the proposed system can robustly measure the position of a football in real time and is effective for producing live football broadcasts.

Keywords Visual object tracking · Machine learning · Projective transformation · Sports-video analysis

1 Introduction

Visual object tracking is important in computer vision because it can be used in various scenarios, such as surveillance, robotics, and human-computer interaction [21, 38]. However, tracking objects in actual video sequences is generally very difficult due to illumination change, occlusion, background clutter, etc.

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Sports-video analysis is one scenario for which visual object tracking is in strong demand [35, 42]. Several systems have been developed that measure the positions of football players, such as TRACAB [43], Deltatre [12], and Piero [31]. In addition to the movement of players, the movement of a ball is very informative because it decides the game. Although most audiences focus on the ball position and its motion, it is still difficult to track a ball with these systems because the size of the object is much smaller than that of players. Ball tracking tends to fail due to noise objects such as the players' shoes and white uniforms. Therefore, a robust system for measuring the position of a ball is needed.

Data on ball positions can be used for various applications, such as calculating statistical data related to a ball and the production of live football broadcasts. For example, real-time ball position data are useful for automatic audio mixing because more than ten mics are usually set around a football field during a live broadcast. The volume of the mics, which are near the ball, can be automatically turned up with the data, and the sound of the ball being kicked can be clearly captured.

The coordinates of the ball position also can be used for displaying the position of a ball in real time from a top view as an internet service. Although there are several services that display the ball and players' positions by using the TRACAB system [26], doing so accurately is still difficult. We thus focused on tracking a ball from football video in this paper.

Several visual object-tracking systems have been proposed that are based on appearance models of targets [9, 22, 33]. Although these systems are effective, they often include complex appearance models, so keeping up with the 30 frames per sec produced by many modern cameras is a struggle. In addition, they were evaluated with video datasets that contain only several hundreds of frames. It is difficult to track targets for a long time with these systems. A practical system that can be used for live professional football games must include real-time processing and robust tracking for an entire game as well as high tracking accuracy.

The versatility of the systems is also important. Professional football games are held at various stadiums and various times both day and night. Although the differences in appearance between stadiums and changes in illumination throughout the day can prevent the systems from robustly tracking a ball, the ball's position should be robustly measured regardless of these environmental differences.

We developed a robust system for measuring the position of a ball that can be used for football games. It robustly tracks a ball object from fixed multi-view cameras installed at football stadiums and outputs the ball's position with top-view plain coordinates in real time. The movement of a ball is different from other objects, such as players. We thus focused on the motion features of a ball object in addition to its appearance features [40]. Tracking results from multiple cameras are integrated in consideration of the reliability of these results, and the final ball position is accurately measured.

The proposed system was evaluated with videos shot during four professional football games at multiple stadiums during both the day and night. The experimental results indicate that the system can robustly detect and track a ball object in real time by integrating tracking results from four fixed cameras and robustly measure the ball position even at different stadiums and times. The results of a demonstration experiment proved that real-time ball data are useful for automatic audio mixing in the production of live football broadcasts.

The remainder of this paper is organized as follows. We discuss related work in Section 2, give an overview of the proposed system in Section 3, explain modules of the system in Section 4, present experimental results in Section 5, and conclude the paper in Section 6.

2 Related work

Visual object tracking is important, and many systems have been proposed [21, 38]. It is also important for sports-video analysis because the positions of players and the ball attract much interest from spectators. Many systems have been proposed that track people and recognize their actions from camera images [23, 24, 37, 44]. However, the computational cost of these systems tends to be high because of their complicated calculation. Real-time computation is required for systems that are used in live sports broadcasts.

Due to the popularity of football, several systems, such as TRACAB [43], Deltatre [12], and Piero [31], have been developed that measure players' positions by tracking them. Although these systems can create a large amount of statistical data related to players' positions, it is still difficult to automatically track a ball because the object is much smaller than players.

Visual object tracking is basically performed by matching the representation of a target model built from the previous frame of video. Most representations are made from the appearance features of target objects [6]. A variety of appearance features have been proposed, such as scale-invariant feature transform (SIFT) [27], speeded up robust features (SURF) [3], oriented FAST and rotated BRIEF (ORB) [34], and local binary pattern (LBP) [30]. These features have been used to solve many problems in computer vision and are used to train discriminative classifiers for visual object tracking. In general, effective feature representations have highly dimensional and complicated models, so they tend to incur a high computational cost.

Recently, online machine learning methods for object tracking based on discriminative classification have been proposed [1, 11, 14, 15, 19]. They build a classifier to distinguish target pixels from background pixels and update the classifier by extracting new samples in the next frame. The boosting method tracks objects by updating its classifier every frame, extracting positive samples from the target region and negative samples from outside it [15]. The multiple instance learning (MIL) tracker is another online machine learning method [1]. It shows improved robustness to incorrectly labelled training samples. The tracking learning detection (TLD) tracker simultaneously detects a target object in a video, learns its appearance, and tracks it [19]. It estimates its detector's errors and updates it to avoid these errors in the future. Correlation filter based trackers have been regarded as state-of-the-art techniques for object tracking [5, 8, 17, 41]. The kernelized correlation filter (KCF) is a novel tracking framework that utilizes the properties of the circulant matrix to enhance the processing speed [5].

Although discriminative classifiers robustly track targets by updating target detectors, most of their models are created from a target's appearance, so it is difficult to accurately track small ball objects that contain few visual features with such classifiers. Moreover, it is difficult to re-acquire target objects after tracking failures with the classifiers because the appearance model would have been changed by being updated. The motion features of a target are not often used for tracking despite being an important cue.

The keypoint trajectory has been regarded as a useful motion feature [7, 28, 32, 45]. It contains the positional history of a keypoint. The local invariant feature tracks (LIFT) descriptor is a frequency histogram of the motion-vector directions in a keypoint trajectory [28] and is mainly used for human-motion recognition. The feature is invariant for the duration of a trajectory because its dimensions are fixed. However, it cuts off the temporal information from a trajectory. The proposed system thus extracts a variety of

features from a ball trajectory, including direction and velocity, and uses them for ball tracking.

A ball tends to be hidden by players, and ball tracking is also interrupted due to the detection of other noise objects. Thus, estimating a hidden true value from system observation, which contains missing data and measurement errors, is another problem. The Kalman filter operates recursively on streams of noisy input data to produce a statistically optimal estimate of the underlying system state [20]. The particle filter also estimates a hidden true value by integrating weak estimation calculated with many particles [13]. Although it can estimate the non-linear and non-Gaussian motion of a target, it tends to incur a high computational cost, which is unsuitable for live sports-video analysis. Our proposed system instantly estimates the true position of a ball from multiple observations that contain errors in consideration of the reliability of tracking results.

Multi-task multi-view learning is another important topic for improving the performance of a measurement system [25]. Nieto et al. proposed a system that measures a player's 3-D position in tennis games by using multiple cameras [29]. The system calculates the position of the player by averaging the results from the multiple cameras. Thus, a specific camera with a tracking error tends to cause a large measurement error. In our system, tracking results from each camera are effectively integrated by considering the reliability of the ball positions of each camera, so the final ball position is accurately measured. Moreover, some multi-camera systems that measure ball positions in actual games have been used in practice [16, 39]. However, these systems require a huge number of cameras and strict camera calibration. Few devices and simple operation are desirable for systems that are used for live sports broadcasts.

3 System overview

An overall flow chart of our proposed ball-position-measurement system is shown in Fig. 1. The system consists of two main processes. The first is ball detection and tracking, and the other is integration of tracking results. The system uses several consumer HD cameras, the same number of ball-tracking PCs, and a ball-position-integration PC. All PCs in the system are common consumer PCs. Unlike conventional player-tracking systems such as TRACAB [43], our system does not require any special equipment, so it can be used at any football stadium. This is one advantage of our proposed system.

We define N_C as the number of cameras. N_C fixed cameras are set at a football stadium, and video sequences from them are input to the ball-tracking PCs. Multiple cameras are normally required because it is difficult to shoot an entire football field by using a single camera. The ball-tracking processes for all cameras are executed in parallel. The detected ball positions are input to the integration PC via Ethernet, and a final ball position is measured.

Although ball positions are measured with the image coordinates of each camera, the final position is presented with top-view plain coordinates, which ignore the height axis in real 3D space, as shown in the right image in Fig. 1. We denote these coordinates as "top-view coordinates." The unit of theses coordinates is in meters [m].

The system can be started by simple calibration. We only need to prepare a homography matrix for each camera. These matrices are calculated by simply setting more than four corner points on the football field (see Subsection 4.3). This system thus does not require any complicated camera calibration, such as measuring the focus length, position, and orientation of cameras, because the transformation from camera image to top-view coordinates is 2D,

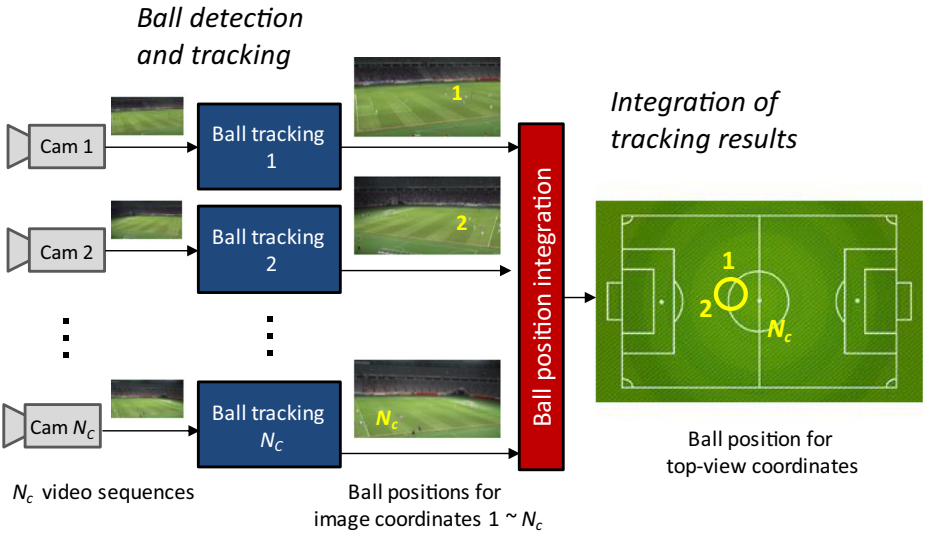

Fig. 1 Overall flow chart of our proposed system. Balls are tracked in parallel, and their positions with camera image coordinates are output (yellow numbers in middle figures). Ball positions from multiple cameras are integrated, and top-view coordinates of final ball position are measured (yellow circle in right figure)

which ignores the height axis of the ball. Once the matrices are calculated, ball tracking starts, and the ball position is automatically measured without any manual operation. Many visual-object-tracking systems require a manual restart when they miss target objects because the appearance models have been updated. However, the proposed system, which uses both appearance and motion features, automatically re-tracks a ball object so that the ball position is measured throughout the game.

Four professional football games in Japan were recorded by using $Nc = 4$, and the videos were used to develop and evaluate our system.

4 Ball-tracking and position-estimation method of proposed system

4.1 Flow of ball-tracking and position-estimation method

As mentioned above, the ball-tracking and position-estimation method of our proposed system consists of two main processes, a detection-and-tracking process and tracking-result-integration process. The former detects and tracks a ball in the video sequences of each camera in parallel and outputs a ball position for each image coordinate. The latter integrates the tracking results from the former processes and estimates the top-view coordinates of the final ball position.

Figure 2 shows the flow of this method. We denote the initial and final ball positions as P_I^c and P_T, respectively, where I and T denote the identifier between image coordinates and top-view coordinates, and c ($c \in N_C$) denotes the camera number. The component of each process corresponds to the processes in Fig. 1.

Our system sequentially detects and tracks a ball on the basis of its appearance and motion. First, ball-candidate objects are eliminated with appearance-based features and then eliminated

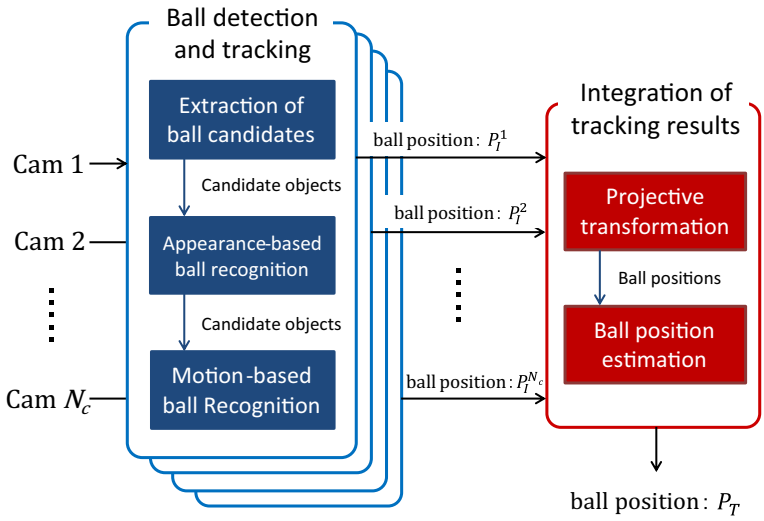

Fig. 2 Flow of ball-tracking and position-estimation method of proposed system. Component of each process corresponds to PCs in Fig. 1

again with motion-based features. Thus, only ball candidates that have both a ball-like appearance and ball-like motion are detected.

As the flow shows, the proposed system estimates P_T by integrating P_I^cs from multiple processes. The ball position is measured in real time because all ball-tracking processes, which are executed in parallel, are executed within 33 msec (30 fps). This distributed processing architecture enables both robust estimation and real-time computing for ball-position measurement. In the following section, we explain these two processes and their modules in detail.

4.2 Ball detection and tracking

4.2.1 Ball-candidate-extraction module

In this module, the proposed system extracts candidate ball objects on a football field by using low-level image features. This module plays a significant role in speeding up the entire ball-detection-and-tracking process because the features in the following subsections incur a relatively high computational cost.

Figure 3 shows the processing flow of this module. First, the module calculates frame differences by subtracting the brightness of each pixel between two temporally consequent frames. This calculation is limited to a football field area to narrow the ball-search area, as shown in the black region in Fig. 4.

Neighboring pixels that have high frame differences are connected, creating blobs, and these blobs are regarded as moving objects. Several moving objects are shown on the field area

Fig. 3 Flow of ball-candidate-extraction module

Fig. 4 Sample of moving objects. Black region: field area. Objects on black region: moving objects including true ball (indicated with red arrow)

in Fig. 4. These objects are extracted on the basis of the relative illumination change between the consequent frames instead of the absolute brightness. Thus, this module can be applied to both day and night games, regardless of the changes in brightness.

Then, the module roughly narrows down the number of ball candidates on the basis of low-level image features: color (white in RGB color space), size (number of pixels in blob), and shape (circular level). The thresholds of this filtering process are set with wide enough margins, taking care not to miss a true ball object. A true ball object is indicated with the red arrow in Fig. 4. Although a wide range of candidate objects including the true ball object are extracted at this stage, the object is effectively detected with the following two types of ball-recognition modules.

4.2.2 Appearance-based ball-recognition module

Whether candidate objects are identified as balls is determined on the basis of their appearance. A supervised machine-learning framework is used for this module. Figure 5 shows the processing flow.

In the training phase, thousands of ball images and the same number of other images are manually cut out from football video sequences beforehand. They are used as positive and negative samples to adapt a supervised machine learning method. Figure 6 shows samples of positive and negative images. A ball object at the position nearest the camera is larger than that

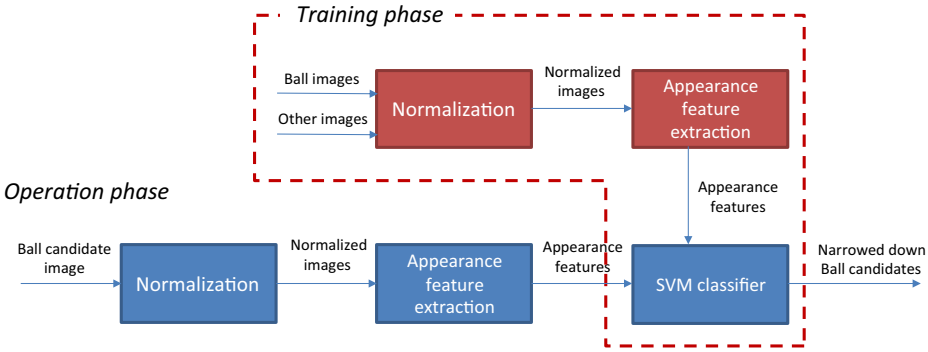

Fig. 5 Flow of appearance-based ball-recognition module

at the farthest position, so their sizes are automatically determined in the range of 8 × 8 to 24 × 24 pixels depending on their extracted position. The color and brightness are also normalized for each extracted image assuming illumination changes during day games.

A combination of an RGB color histogram [36] and LBP [30] features are used as the appearance features of our system. The RGB color space is segmented into 3 × 3 × 3 blocks, and a 27-bin color histogram is created by counting pixel colors in an image. The histogram is normalized by the image size. An LBP feature, which is a gradient-based image feature, is also created from the same image. Although the number of dimensions of the feature is usually 256, we use a 16-dimensional LBP to reduce the computational cost. As a result, the total number of dimensions of the appearance features is 43 (= 27 color histogram +16 LBP). The number of dimensions is smaller than that of many conventional appearance features, such as SIFT and SURF.

A binary support vector machine (SVM) classifier [10] is then created by learning the above appearance features from training samples. The Gaussian kernel is used for this. The training can be carried out in a few minutes due to the small number of dimensions of the features.

In the operation phase, small regions around candidate objects are automatically extracted, and the color histogram and low-dimensional LBP features are calculated in the same manner as the training samples. The classifier determines whether each region can be regarded as a ball on the basis of the appearance features, and the number of ball candidates from the ball-candidate-extraction module is narrowed down.

4.2.3 Motion-based ball-recognition module

Several candidate objects are acquired by using the above-mentioned module at each frame. In this module, the candidate objects are tracked to create their trajectories. Then, the module selects a ball object on the basis of features from its trajectory.

First, two candidate objects at different frames are matched. The Bhattacharyya distance of the color histogram and LBP, which are used for appearance-based ball recognition, is used to calculate the distance between two candidate objects [4]. If the distance is small, the two objects are connected, and a trajectory is created.

A trajectory $\boldsymbol{T}_t^i$ is the positional history of a candidate object. Note that i is the ID number of a candidate object that is being tracked, and t is the frame number. A $\boldsymbol{T}_t^i$ contains a time series of positions detected on camera-image coordinates and is created when two candidate objects at different frames are matched. It disappears when the latest candidate object on the $\boldsymbol{T}_t^i$ does not match any other candidates for a specific number of frames. Its value was experimentally set at 30, i.e., 1 s. This means that a $\boldsymbol{T}_t^i$ exists while a candidate object is visible. After a $\boldsymbol{T}_t^i$

Fig. 6 Positive and negative samples of ball object. Upper: positive samples. Lower: negative samples

disappears, a new ID $t+1$ is tagged on a new $\boldsymbol{T}_t^{i+1}$, even if both trajectories are generated from the same candidate object.

Once the $\boldsymbol{T}_t^i$ is generated, the motion features related to it are calculated. This module creates six motion features from a $\boldsymbol{T}_t^i$:

- Temporal length t^i: duration for which a $\boldsymbol{T}_t^i$ exists [frame].
- Trajectory length l_T^i: total travel distance of a $\boldsymbol{T}_t^i$ for top-view coordinates [m].
- Velocity v_T^i: average velocity of a candidate object for top-view coordinates [m/frame].
- Direction histogram $\boldsymbol{h}_I^i$: 9-bin histogram of directional frequency of motion vectors in a $\boldsymbol{T}_t^i$.
- Magnitude of motion vectors m_I^i: average magnitude of motion vectors in a $\boldsymbol{T}_t^i$ for image coordinates [pixel].
- Standard deviation of magnitude s_I^i: standard deviation of m_I^is in a $\boldsymbol{T}_t^i$ for image coordinates [pixel].

Here, i denotes the ID number of trajectories, and T and I are identifiers between top-view coordinates and image coordinates, respectively. The features of the top-view coordinates are calculated by projecting the positions of the image coordinates onto the top-view coordinates by using projective transformation. This computation is explained in Subsection 4.3. These features are not affected by the difference in cameras because they have common top-view coordinates.

When the module creates a direction histogram $\boldsymbol{h}_I^i$, a $\boldsymbol{T}_t^i$ is separated into motion vectors. We divided the motion vectors into nine different directions including magnitude zero, as shown in Fig. 7. The frequency histogram was normalized by t^i so that the summation of all elements in it is always 1.0. The average m_I^i and its s_I^i were also calculated from motion vectors in the $\boldsymbol{T}_t^i$.

The supervised machine-learning framework was used to train a classifier of trajectories. The classifier of an SVM regression was trained with positive and negative samples, which were manually extracted. The classifier outputs the possibility of a ball trajectory as a continuous value. Thus, this module can select a unique trajectory from several candidates by sorting their scores. The latest ball position, i.e., P_I^c, on the selected trajectory is output as the result of camera c. Figure 8 shows an example processing image during ball tracking.

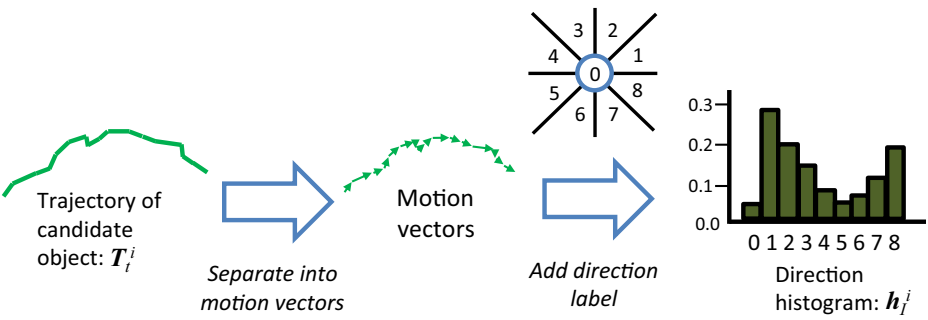

Fig. 7 Creation of direction histogram from trajectory. Trajectory is separated into motion vectors, and direction histogram is created by counting direction labels of each vector

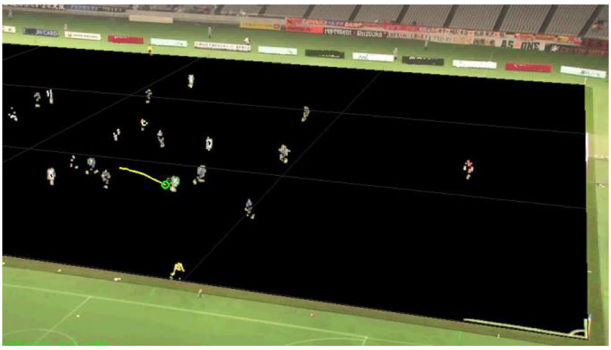

Fig. 8 Processing image during ball tracking from video sequence of specific camera

The green circle in the image denotes the position of the ball, and the yellow curve denotes its trajectory.

4.2.4 Reliability of detected ball position

A ball object in football video is frequently hidden behind players. Although the system cannot detect a ball object when it is occluded, an expected position should be output, even though the ball is hidden. The proposed system thus regards the nearest object from the latest ball position as the ball in the frame when no ball objects are detected.

The elapsed time E^c [frame] is measured while ball detection fails and is output with the expected ball position, i.e., P_I^c. To be precise, P_I^c denotes the latest ball position on a selected $\boldsymbol{T}_t^i$ when $E^c = 0$ (when ball tracking is successful) and denotes the position of the nearest object from the latest ball position when $E^c > 0$ (when ball detection fails). The values are calculated at every frame for each camera in parallel. The results from all cameras are effectively integrated, and P_T is estimated in the following process.

4.3 Integration of tracking results

In this process, the P_I^c values acquired from multiple cameras ($c \in N_c$) are integrated on the top-view coordinates, and P_T is finally estimated. The overview of this process is shown in Fig. 9. The process consists of two modules: projective transformation and ball-position estimation. P_I^c is transformed on the top-view coordinates P_T^c, and they are integrated into P_T considering their reliability E^c. We explain these modules in detail in the following subsections.

4.3.1 Projective-transformation module

This module projects the P_I^c of image coordinates ($c \in N_c$) onto top-view coordinates P_T^c. The projection is executed by using projective transformation [2], which is the projection from a planar surface to another. The top-view coordinates are also 2D coordinates produced by dropping the height component of real 3D coordinates. Therefore, the camera-image coordinates of a ball position can be projected onto top-view coordinates by projective transformation.

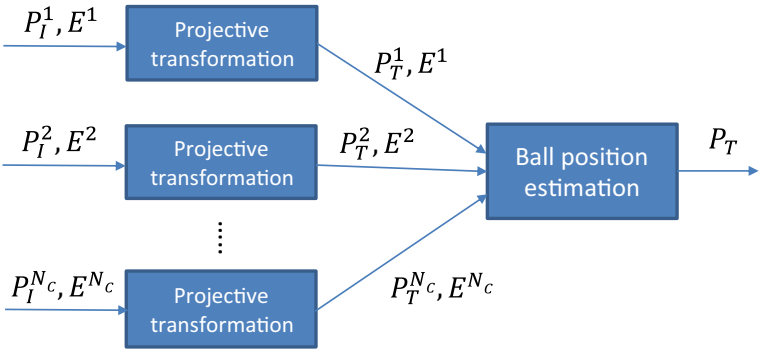

Fig. 9 Flow of process for integrating tracking results

Figure 10 shows a sample projective transformation from a camera image to a top-view image. The right side of the football field shot with a single camera was translated to almost half the area in the top-view image.

The P_I^c from all cameras ($c \in N_C$) are projected onto top view coordinates by using the transformation

$$P_T^c = h\left(P_I^c\right) \qquad (c \in N_C), \tag{1}$$

where $h(\cdot)$ is a projective-transform function [2]. All ball positions from different cameras are projected onto the same planar surface.

In practice, an operator of the proposed system must set more than four pairs of corresponding points between camera-image coordinates and top-view coordinates, and a homography matrix for each camera should be prepared before the game. This is a single operation done during system calibration before a game because any points for camera-image coordinates can be projected onto top-view coordinates by using these homography matrices.

4.3.2 Ball-position-estimation module

In this module, the P_T^c of all cameras for the top-view coordinates ($c \in N_C$) are integrated, and P_T is estimated. The ball position is robustly estimated in consideration of E^c by linear summation.

$$P_T = \frac{\sum_C \left\{w^c P_T^c\right\}}{\sum_C w^c}, \left(w^c = M - E^c\right), \tag{2}$$

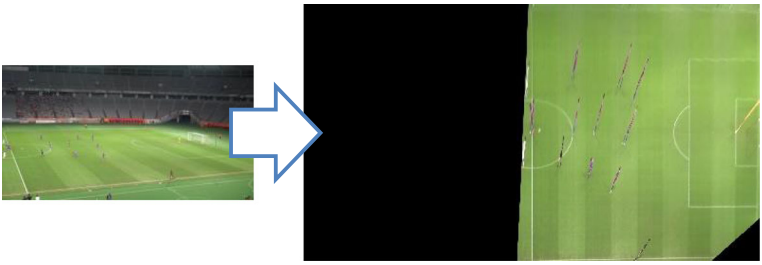

Fig. 10 Sample projective transformation from camera image coordinates to top-view coordinates. Camera image shot at right side of football field is transformed to almost right half of image in top-view coordinates

where w^c denotes the weight of camera c and is determined by subtracting E^c and constant value M. We set $M = 150$ frames, i.e., 5 s. The P_T is calculated in real time because this equation is simple and its computational cost is low. Even if a ball is occluded by a player in a specific camera image, the final ball position is accurately measured because the weight of the ball position from the occluded camera tends to be low.

Figure 11 shows a sample integration of ball positions. The numbers and their positions on the image show c ($c \in N_C$) and the detected ball position of each camera. The brightness of the numbers denotes the w^c (yellow: high, black: low). This means that the ball position with the black number had a long elapsed time from the frame when ball detection failed. The yellow circle shows the final estimated position of the ball, and the red point denotes the true ball position, which was manually annotated. The image shows that the proposed system precisely estimated the ball position, even though the detected positions of some of the cameras were far from the true position. The ball positions were effectively integrated by calculating their weights in this module. This integrative computation scheme also makes the proposed system more versatile. A ball can be robustly tracked even at different stadiums and different times.

5 Experiments

5.1 Experimental conditions

We conducted experiments to confirm the effectiveness of the proposed system. Four professional football games held at two different stadiums in Japan during both the day and night were recorded by using four fixed cameras and were used for an evaluation, as mentioned in Section 3.

Four consumer HD video cameras were set on spectator seats, as shown in Fig. 12. They were fixed with a tripod and recorded the games without any manual operation. Two cameras covered

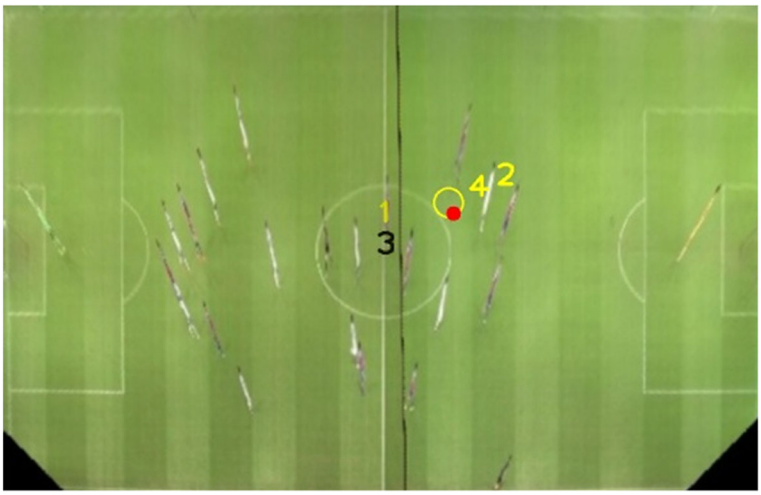

Fig. 11 Sample ball-position integration. Numbers: ball positions detected from each camera (brightness denotes weights). Yellow circle: final estimated ball position. Red point: true position of ball that was manually annotated

one side of the football field, as shown in Fig. 13. They were positioned far from each other to shoot from different fields of view to avoid as many occlusions as possible.

All four cameras recorded the entirety of the four games. Thus, 24 h (= 45 min × 2 halves × 4 games × 4 cameras) of video sequences were recorded. Although conventional visual-object-tracking systems are typically evaluated with small datasets, we created this large football-video dataset to evaluate the effectiveness of the proposed system for practical use.

The first halves of the four games were used for training, and the second halves were used for the evaluation. The ball positions every 10 frames (3 fps) in the entire dataset were manually annotated for all cameras and used as the ground truth for the ball positions.

5.2 Appearance-based ball recognition

We first evaluated the accuracy of the appearance-based ball-recognition module described in Subsection 4.2.2. We prepared 4500 images of balls by extracting regions at manually annotated ball positions in the first halves of the two night games for the four cameras. The same number of negative samples, i.e., regions not containing a ball, was also prepared. The appearance-based ball-recognition feature was calculated from all positive and negative samples, and a binary SVM classifier was trained in a supervised-machine-learning manner. Four thousand five hundred images, the same number as training images, of positive and negative samples were also extracted from the second halves of the two games and used for the evaluation.

Many conventional appearance features, which have been used for many applications in computer vision, were used for comparison with the proposed feature: moment [18], color histogram [36], LBP [30], normalized cross correlation (NCC) [6], ORB [34], SURF [3], and SIFT [27].

We evaluated the appearance-based ball-recognition feature in terms of both accuracy and speed. Table 1 lists the results. The accuracies of the color histogram and LBP were high, and the proposed system, which combines both features, achieved the best accuracy. Although the system used a 16-dimensional LBP feature, its accuracy was better than that of a 256-dimensional LBP feature. The combination of two different types of features, color-based and gradient-based, improved robustness and provided the best results.

The calculation speed of the appearance-based ball-recognition feature of the proposed system was 0.66 msec, which was worse than that of the moment feature. Although the speed of the appearance-based ball-recognition feature was not the best, it was faster than most of the other features. Reducing the dimension of the LBP feature contributed to this fast computation. In

Fig. 12 Cameras set on spectator seats (Left: day game. Right: night game)

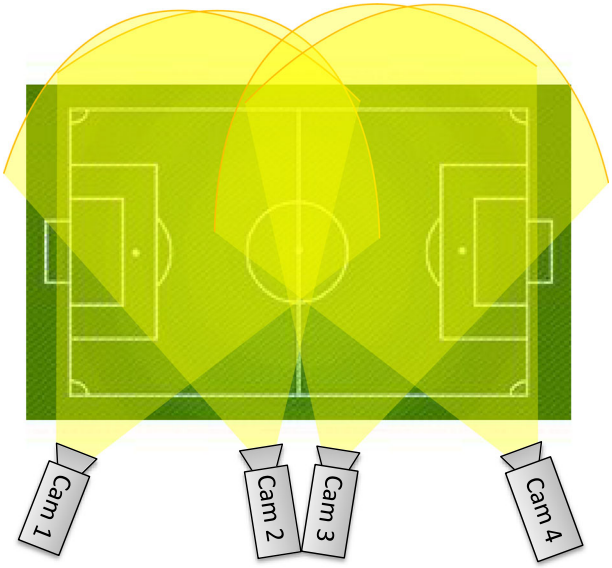

Fig. 13 Positions of four cameras and their fields of view

addition, accuracy should prioritized over speed as long as the feature is processed in real time, i.e., within 33 msec.

Even though the number of dimensions of ORB, SURF, and SIFT is high, their accuracies were low because the size of the images was small. The images of the candidate objects were in the range from 8 × 8 to 24 × 24. ORB, SURF, and SIFT are descriptors of local keypoints, so they could not extract sufficient numbers of keypoints in such small regions. Moreover, the computational speeds of ORB, SURF, and SIFT were slow because their dimensions are high and their calculations are complicated. The results show that these features are not suitable for detecting a small ball object in real time.

We conclude that this feature of our proposed system is the best for appearance-based ball recognition. Its high accuracy and low computational cost contributed to the high performance of the proposed system during real-time operation.

Table 1 Results of appearance-based ball recognition

	Dimension	Precision	Recall	F-Measure	Speed [msec/frame]
Moment [18]	10	0.49	0.87	0.63	0.49
Color histogram [36]	27	0.86	0.91	0.86	0.51
LBP ($P = 8$) [30]	256	0.79	0.84	0.81	0.87
NCC [6]	—	0.95	0.64	0.76	1.24
ORB [34]	32	0.64	0.99	0.78	2.97
SURF [3]	64	0.79	0.66	0.72	9.36
SIFT [27]	128	0.61	0.99	0.76	38.15
Proposed	430	87	0.94	0.91	0.66

5.3 Motion-based ball recognition

Next, we evaluated the accuracy of the motion-based ball-recognition feature of the proposed system. The trajectories at the true ball positions, which were manually annotated, were extracted as positive samples, and other trajectories were extracted as negative samples. One thousand five hundred positive samples and 3000 negative samples were extracted from the training dataset, i.e., video sequences of the first halves of the two night games with the four cameras. The classifier of an SVM regression was created with the six motion features mentioned in Subsection 4.2.3 from the samples. The same 4500 trajectories were extracted from the test dataset and used to evaluate the accuracy of the motion-based ball-recognition feature.

A velocity-based classifier and LIFT-based SVM classifier were used as the conventional methods. The velocity-based classifier simply identified whether a trajectory was positive or negative depending on its velocity. The LIFT-based SVM classifier was trained with LIFT features [28] that were extracted from the same trajectory samples.

The results are listed in Table 2. The motion-based ball-recognition feature performed the best, proving that the six features acquired from a trajectory are effective for detecting a ball trajectory in football video. In addition, these trajectory features are normalized for the common top-view coordinates, so they were not affected by the difference in cameras. These normalized features between cameras contributed to the high accuracy of the proposed system.

5.4 Comparison with state-of-the-art trackers

Next, we evaluated the performance of our proposed tracking method. We compared it with state-of-the-art tracking methods, which use online machine learning, by using the criteria of accuracy and processing speed. We used TLD [19], MIL [1], and KCF [17] as methods for comparison.

Video sequences of 900 consecutive frames were used for this evaluation. The initial position of a ball, which was common for all trackers, was manually set, and each method individually tracked a ball until the last frame. We did not reset their tracking position even if they lost the ball. Error distances between the manually annotated ground truth and estimated positions in the image coordinates were measured for each tracking method, and their average values were calculated. The average elapsed times for one cycle of tracking process for all tracking methods were also measured. The results are shown in Table 3.

The proposed method achieved the best accuracy among all of the trackers. Other online tracking methods tended to lose the ball when the ball was occluded by players. When a ball object was occluded by a player, inadequate features were extracted from the region. Then, the methods incorrectly updated their classifiers and lost the ball object. Our proposed ball tracker accurately re-tracked a ball after occlusion because it detected a ball by using a fixed classifier that was trained with large numbers of ball images beforehand.

Table 2 Results of motion-based ball recognition

	Precision	Recall	F-Measure
Velocity-based	0.90	0.55	0.68
LIFT [28]	0.63	0.99	0.77
Proposed	0.75	0.95	0.84

Table 3 Comparison with other tracking methods

	TLD [19]	MIL [1]	KCF [17]	Proposed
Error [pixel]	318.9	86.6	41.6	25.5
Speed [msec/frame]	106.2	42.1	5.6	0.66

Our proposed ball tracker also achieved the best speed. Although other online trackers update their classifiers every frame, our ball tracker never does so during ball tracking. This caused the computation of our method to be fast. An image processing system that is used for broadcasting is required to process within 33 msec to keep up with the 30 frames per second produced by broadcast cameras. These evaluation results indicate that our proposed method is the best approach for use in live broadcasts.

5.5 Integration of ball position from multi-view cameras

We then evaluated the accuracy of estimating the position of a ball with the proposed system. The system outputs ball positions of top-view coordinates every frame. We measured three types of indices for this evaluation.

The first is the average error distance, which is the distance of the top-view coordinates between the estimated and true ball positions manually annotated. The second is the time rate within 10 m, which is the time rate at which a ball position is estimated within 10 m from the true ball position in an entire test sequence. The third is the computational speed, which is the average elapsed time until tracking results at a frame are integrated and a ball position is estimated.

Ninety minutes of video sequences were used for this evaluation, containing the second halves of the two night games (= 45 min × 2 games). The ball positions detected with each camera, which were tracked with our proposed system, were used as the input data of this experiment.

The accuracies of the Kalman filter [20] and particle filter [13], which are conventional methods for estimating a hidden true value, were compared with the ball position-estimation method of the proposed system. These methods track objects by predicting their next positions on the basis of motion dynamics. Linear uniform motion dynamics and 1000 particles were used as their parameters in this experiment.

Table 4 lists the evaluation results. A linear Gaussian model for object tracking was assumed with the Kalman filter, so it was difficult to track a football that moved non-linearly. A non-linear non-Gaussian model was assumed with the particle filter, so it was relatively robust for tracking a ball that moved with unpredictable dynamics. However, it required a large number of particles to accurately track a ball. This led to a high computational cost, so the elapsed time of the particle filter was longer than that of the other methods.

Table 4 Results of ball-position estimation

	Average error distance [m]	Time rate within 10-m error [%]	Speed [msec/frame]
Kalman filter [20]	8.2	74.7	15.6
Particle filter [13]	6.5	81.7	28.1
Proposed	5.3	87.4	15.7

The average error distance of the ball position-estimation method of the proposed system was 5.3 m, and the time rate for which the error distance was within 10 m was 87.4%. These results were better than those of the other two methods. The results for the integration process of tracking contributed to this high accuracy. The ball positions detected from the multiple cameras were effectively integrated in consideration of the cameras' reliabilities. Even a ball that was hidden behind players in a video that was shot with a specific camera could be detected from another video that was shot with another camera from a different angle with high reliability. The scheme also prevented our system from being interrupted by observation errors, which were caused by misdetection of noise objects, contributing to robust estimation.

5.6 Robustness under different conditions

We also evaluated the robustness of the proposed system under different conditions by using a video dataset of multiple stadiums and different times. Four types of average error distances and a time rate within an error of 10 m were calculated with different training and test data. As shown in Table 5, these data were created by changing stadiums and time.

The first row corresponds to the results presented in Subsection 5.4, which were obtained during the two night games. The second row shows the results of the day games, which were worse than those of the night games. The illumination changed during the day games, and shadows of the stadiums and players appeared, as shown in Fig. 14. These shadows prevented the system from accurately detecting the ball. The third row shows the results obtained during the day games by using the classifier trained at different stadiums. The grass color slightly changed between the stadiums, and the camera positions also changed depending on the stadiums. These differences made accurate tracking difficult. The fourth row shows the results obtained during the day games by using the classifier trained during the night games. The results were much worse than those in the first row. The system could not create a proper classifier for the day games from the night-game data because of the illumination and color differences between day and night.

To summarize the results, accurate tracking is relatively difficult with the proposed system when it uses a classifier trained at different game times. However, the differences between stadiums were not drastic, so we need to prepare only two types of classifiers for day and night games trained at multiple stadiums. Thus, the proposed system can robustly track a ball even during the day and night and at different stadiums by using a proper classifier.

5.7 Demonstration experiment

Finally, we conducted a demonstration experiment of our system for automatic audio mixing during a real live football broadcast of the Japan professional football league on

Table 5 Average error distances and time rates within 10-m error under different conditions

Training data		Test data		Average error distance [m]	Time rate within 10 m error [%]
Stadium	Day time/ Night time	Stadium	Day time/ Night time		
A	Night	A	Night	5.3	87.4
B	Day	B	Day	5.7	80.6
B	Day	A	Day	6.9	76.0
A	Night	B	Day	8.1	69.1

Fig. 14 Sample of top-view image during day game. Shadows of stadium and players can be seen

May 30th in 2015. The real-time ball position measured with our system was used for input to the audio mixing system. Twelve mics were installed around the football field, and the volume of the mics, which were near the ball, was automatically turned up by referring to our ball-position data. The height of the ball is not required for automatic audio mixing because the mics can capture a wide range of sounds. A photograph of our system during this experiment is shown in Fig. 15.

Our system, which was trained with other two night games, robustly measured ball positions throughout the entire 90-min game in real time. Table 6 shows its tracking accuracy. The average error between true ball position and estimated ball position was 6.1 m, and the time rate within a 15-m error distance exceeded 90% for the entire game. The audio-mixing system adequately worked with the ball-position data from our system, and ball kicking sounds were clearly captured. Our system contributed to making audio mixing efficient and effective. We thus confirmed that our system is useful in the production of live football broadcasts.

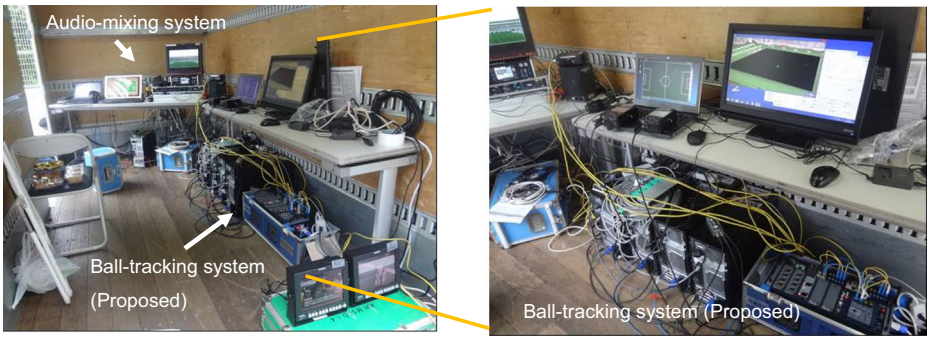

Fig. 15 Photograph of our system during demonstration experiment

Table 6 Average error distances and time rates within 10- to 15-m error in demonstration experiment

Time rate within X-m error			Average error
10 m	12 m	15 m	
83%	88%	92%	6.1 m

6 Conclusion

We proposed a ball-position measurement system that can be used for football games. Video shot with multi-view cameras at a football stadium was used for input. The system uses two discriminate features for detecting and tracking a ball; one is for its appearance, and the other is for its movement. Tracking results from multiple cameras are integrated in consideration of the reliability of these results, and the final ball position is measured by using top-view plain coordinates of the football field. The experimental results of video sequences of four professional football games proved that the appearance and motion features of the proposed system can robustly detect and track ball objects in real time. Tracking results from multiple cameras were effectively integrated, and the final ball position was accurately estimated. We also confirmed that the system is robust against environmental differences when using a proper classifier. The results of the demonstration experiment also proved that real-time ball data, which were obtained with our proposed system, are useful for automatic audio mixing during the production of live football broadcasts.

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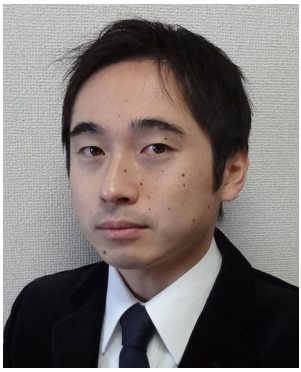

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